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OmEGa(Ω): Ontology-based information extraction framework for constructing task-centric knowledge graph from manufacturing documents with large language model

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ABSTRACT

Manufacturing industry relies heavily on technical documents that encapsulate specialized knowledge essential for optimizing production and maintenance processes. However, extracting meaningful insights from these documents is challenging due to their complex structure, domain-specific terminology, and multimodal content, which includes text, images, and tables. Furthermore, there is a contextual gap between the generic training data of pre-trained language models (PLMs) and the specialized knowledge required for manufacturing documents. To address these issues, a Task-Centric Ontology (TCO) is designed to describe fundamental manufacturing tasks, and develop OmEGa, an Ontology-based Information Extraction Framework for Task-Centric Knowledge Graphs. OmEGa leverages large language models (LLMs) to perform instance recognition and relation classification on multimodal documents. By utilizing spatial embedding and modality linking, OmEGa addresses structural challenges, while TCO-driven reasoning mitigates contextual challenges. Experimental results demonstrate the effectiveness of OmEGa, achieving strong performance on both proprietary and open-source datasets. Additionally, a Knowledge Graph Question Answering (KGQA) system built on the extracted task-centric knowledge shows promise in enhancing communication among domain experts in the manufacturing sector.

1. Introduction

Manufacturing process, which aims to transform raw materials into practical products, is fundamental in an industrial field [1]. As the industry advances, enterprises accumulate their own techniques about *how to* improve an effectiveness or performance in the manufacturing and maintenance process. These techniques are documented to facilitate internal connections and interactions among domain experts. As illustrated in Fig. 1(a), such documents usually contain domain-specific terminology requiring advanced expert knowledge, which is unfamiliar to inexperts. Besides, they represent information that is linked to or supplements the body text as images or tables. They are not just manuals but also assemble of critical knowledge reflecting the core technical principles.

To fully leverage this invaluable knowledge, some research attempts to extract information and construct symbolic knowledge, a knowledge graph (KG), from manufacturing documents with natural language processing (NLP). Kumar and Starly [2] utilize a BiLSTM architecture

to perform named entity recognition (NER) in scientific manufacturing process literature. Guan et al. [3] conduct relation extraction (RE) from welding documents with an attention mechanism [4]. In particular, ontology is one of the promising methods to extract information from sparse and specialized documents since it defines domain concepts and rules. There are several studies that design their own ontologies specialized to target domains and utilize them to extract information. For example, Ding et al. [5] handle processing methods with human–cyber–physical fusion, and Wen et al. [6] cover operating conditions for manufacturing.

However, previous studies have overlooked a *task* itself, which is a fundamental activity in the overall manufacturing process. Considering the ultimate purpose of manufacturing documents is to inform ‘task at a specific location to resolve some issue,’ modeling task enables to symbolize the information in manufacturing documents and further to apply at knowledge-grounded systems such as KGQA (KG-based Question Answering).

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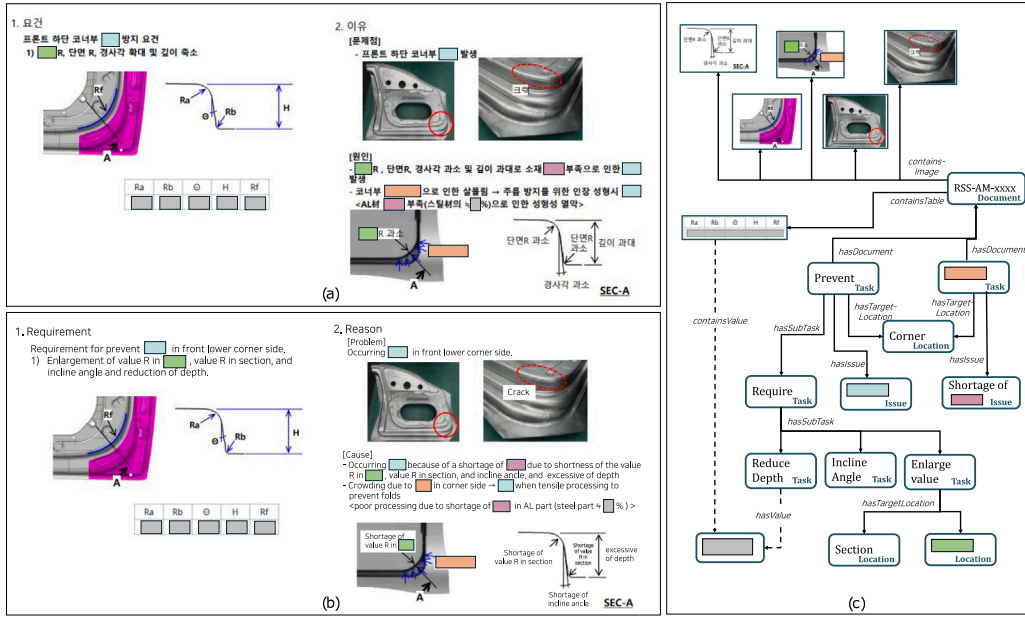


Fig. 1. An example of from the DoorSystem dataset, which is a manufacturing document used in a real-world industry. Some instances and values are blinded due to the security policy of the company. Boxes marked with the same colors indicate the same entities which may be represented into different mentions. (a): An original document written in multiple languages and named RSS-AM-xxxx, (b): English-translated version of (a), and (c): KG constructed from (a).

In this study, we build a Task-Centric ontology (TCO) that defines tasks and task arguments, such as where tasks are performed (location) and what problems they aim to address (issue). An unstructured and domain-specific document, as shown in Fig. 1, is targeted for domain analysis and TCO building. The document is used in real industry, especially the door system process of an automobile manufacturing company.

Meanwhile, there are two main challenges to extracting information from our target data which are distinguished from the prior studies: (1) **structural challenge** and (2) **contextual challenge**.

The **structural challenge** comes from the frequent use of abbreviations and implicit expressions (e.g., DR) rather than concrete words (e.g., door). It is because of insufficient time to make formal descriptions since these documents are drafted urgently to satisfy immediate needs in real-world scenarios. Although they may facilitate communications temporarily, there is no assurance that all members fully understand the documents, even the original author who re-reads after some periods. Furthermore, contrary to the prior studies that only handle textual data [2,3,5–7], there are multimodal contents such as text, table, and image containing domain knowledge in the documents. As in Fig. 1, the relevant information is scattered throughout the various contents, such as the same colored boxes that exist across multiple sentences and images. This leads to a demand for an advanced document-level information-extraction paradigm.

In addition, there is a **contextual challenge** in dealing with domain knowledge. It is mainly caused by a mismatch between training data of PLM, which contains generic domains, and manufacturing documents, which require professional knowledge. Moreover, there is a scarce amount of domain-specific document data for fine-tuning the language models compared to common benchmark datasets. Although Yu et al. [8] and Liu et al. [9] propose post-training PLM with corpus containing terminology, they demand numerous documents. This study considers an absence of a sufficient amount of data for post-training or the resources for annotating them, which is more approximated to real-world industry.

To tackle the above challenges, we propose an **Ontology-based Information Extraction Framework for Task-Centric Knowledge Graphs (OmEGa)** to conduct NER and RE with a pre-trained LLM. It aims to construct a task-centric KG (Fig. 1(c)) from an input document

(Fig. 1(a)), leveraging the TCO and LLM. Augmented training data with a variety of mentions defined in the TCO helps LLM when it encounters unfamiliar contexts to alleviate the contextual challenge. The structural challenges caused by abbreviated expressions is also alleviated. In addition, OmEGa utilizes spatial embedding and modality linking for an adequate understanding of the entire document, resulting in mitigating the structural challenge. To the best of our knowledge, it is the first study to extract domain knowledge from multimodal documents with LLM. We then validate the extracted knowledge fragment through the rules and constraints defined in TCO. The reasoning enables us to expand the missing triples that are not initially extracted and to revise the incorrect extraction. OmEGa achieves remarkable results for instance recognition not only on real-world proprietary dataset (F1 score 71.35%), but also on open-source dataset (F1 score 97.5%), demonstrating its extensibility to the general manufacturing domain. It also reveals its excellence on RE (F1 score 85.9% and 86.6% on the actual and the open dataset, respectively).

Finally, by demonstrating a KGQA system based on the task-centric KG, we show that the proposed framework effectively extracts the task-centric knowledge of manufacturing processes, which in turn improves communication among domain experts.

The main contributions in this study are as follows:

- To model the overall manufacturing engineering domain, we design TCO. The TCO is further used to validate and expand the extracted knowledge to compensate for insufficient extraction.
- We propose **OmEGa** to understand multimodal documents by considering the positions of each component and utilizing a task-centric schema to extract knowledge from input documents.
- We demonstrate the potential of our task-centric KG through various applications, including a KGQA system.

2. Related work

In this section, we introduce existing research on ontology modeling and information extraction specialized in manufacturing or industrial domains. Then the conventional studies on document understanding are summarized.

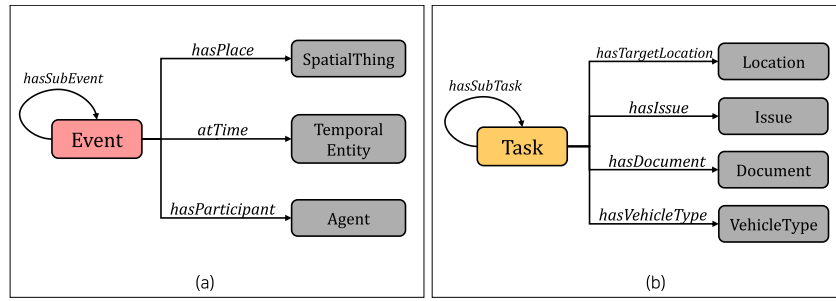


Fig. 2. The representations of Event and Task. (a) the Event class, (b) the Task class derived from (a).

2.1. Manufacturing ontology modeling

A domain ontology, designed by collaboration between ontology engineers and domain experts, represents a schema of specific domain concepts and ensures semantic interoperability and reusability. Because of these benefits, prior studies aim to develop in industrial environments such as manufacturing.

Based on Methontology [10], Zhu et al. [11] construct microdevice assembly ontology, characterizing assembly such as process, product, and numerical design. Wen et al. [6] propose a domain ontology for process knowledge aspect of its condition and equipment and Deng et al. [7] design ontology for a supply chain management. Zhang et al. [12] conceptualize aircraft manufacturing domain, modeling intricate interdependencies to increase predictive accuracy of models employed for quality assessments. In contrast, we developed an ontology for task, a general concept for overall manufacturing process. Moreover, since our target data is specialized to automotive closure systems, partial classes such as the Location models mainly vehicle parts and doors.

2.2. Industrial information extraction

Since industrial documents are typically of uncertain format, recent research attempts to automatically extract information based on specific domain ontologies. Ding et al. [5] classify product manufacturing process into human, cyber and physical forms then used BERT-D'BiGRU-SCRF based on BERT-BiGRU-CRF [13] to extract triple. Wen et al. [6] propose a pattern-based bootstrapping framework with a 2-level masking to extract process knowledge and designed segment rule and feature-based similarity. Deng et al. [7] leverage Generative Adversarial Network (GAN) for active learning of event argument entity recognition, and Dagdelen et al. [14] employ LLM for NER and RE. Moreover, prior studies target only textual documents, which are restrictive to leverage whole information from rich and complex documents.

In contrast, OmEGa extracts instances of task class and non-task class by leveraging label supervised form LLaMA fine-tuning (LS-LLaMA) [15] and factual relations based on the multimodal documents. To complement inadequate extraction due to an awkward domain, reasoning with domain-specific schema is adopted.

2.3. Document understanding

Document AI efforts to parse and understand target documents containing multiple sentences. There are several subtasks for document understanding, such as key information extraction [16–18], natural language interpretation based on tables [19], and question answering for images and tables [20,21] in a given document. Regarding the nature of the document, which includes multimodal components such as tables, images, and texts, and contains arrangement of the components, there are language models specialized to document understanding. LayoutLM [22] is trained on BERT architecture with document data with its image embedding and 2D positional information,

which indicates layout embedding. TILT [23] leverages object detector to extract spatial information. DocFormer [24] proposes multimodal transformer with multimodal self-attention layer. LayoutLMv2 [25] and LayoutLMv3 [26] are enhance version of LayoutLM leveraging improved image embeddings and outperform other document PLMs.

Prior document AIs commonly rely on image embedding such as ResNet [27] or Visual Transformers [28]. In addition, there is barely an attempt to understand domain-specific documents because of their difficulty and scarcity, contrary to this study.

3. Task-centric ontology

An ontology is an engineering artifact which is composed of a specific vocabulary used to describe a certain reality and a set of explicit assertions and rules regarding the intended meaning of the vocabulary [29]. Thus, a domain ontology describes a formal specification of a certain domain, which enables a shared understanding of a domain of interest. Ontologies can generally be divided into two main statements: TBox (Terminological Box) and ABox (Assertional Box). TBox states conceptual components and mainly includes definitions of classes and properties. It serves to set the overall structure by defining concepts and relationships between them. ABox, on the other hand, is an assertion component containing concrete data about actual objects. It states instances based on the concepts defined in TBox and specifies specific relationships between them. TBox and ABox work together to integrate the theoretical structure of a KG with real-world data to enable semantic reasoning. An ontology has often been rebranded as a KG, and it is sometimes used interchangeably. For simplicity of presentation, we use an ontology to refer to TBox and a KG to refer to ABox unless otherwise specified hereafter. In TCO, OWL¹ is exploited to define classes and properties and SWRL [30] is used to express rules.

What to conceptualize is an initial step of the Methontology [10] methodology. As source data, we refer to an accumulated cluster of engineering documents with technical principles, especially for a closure system in an automotive manufacturing company. Although they mainly possess the know-hows aiming for a strong and safe door, we empirically found that the intrinsic purpose of the overall manufacturing document is to inform 'a task at a specific location for some purpose to solve a specific issue'. For instance, a document describes 'a task at vehicle parts or door components for strong doors'. The ontology scope is consequently determined to include the salient classes and attributes that describe the constituent tasks of the manufacturing process.

The proposed ontology, named TCO, inherits an event ontology, which indicates an event occurring in a specific location by an agent, as demonstrated in Fig. 2(a). Since this paper focuses on the task when making doors of a vehicle, the participant and time which are unchanged so omitted as in Fig. 2(b). Instead, what issue to be addressed by a specific task (Issue), what document it is written in (Document), and what vehicle type it is about (VehicleType) are added as task arguments.

¹ <https://www.w3.org/TR/owl2-overview/>

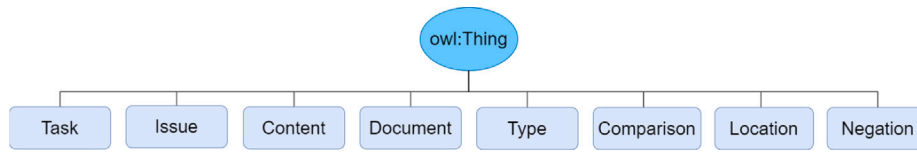


Fig. 3. Top level classes hierarchy of TCO.

Table 1

Subclasses of the task class.

Subclass (level1)	Description	Subclass (level2)	# Subclasses
Apply	Use some object or apply some condition	ApplyStructure, ApplyNotch, ApplyHole, ApplyComponent, ApplyGap, ApplyMarking, ApplyType, ApplyShape, ApplyCondition, ApplyNumeric, ApplySection, ApplyColor, ApplyDirection, ApplyPosition	15
Operate	Use machines to change state	OperateHemming, OperatePiercing, OperateBrazing, OperateMounting, OperateSealing, OperateWelding, OperateForming, OperatePressing, OperateBurring, OperateTrimming, OperateElectroDecomposition	11
Retain	Keep to have some value or state	RetainSection, RetainStrength, RetainSpace, RetainTemperament, RetainValue, RetainMolding, RetainUsability, RetainVisibility	8
Modify	Change some state	Delete, Linearize, Move, Open, A/S, Cover Place, Paint, Encrust, MakeRound, Alter, Mold, Compress, Wire, Install, Assemble, Scrap, Sign, Insert, Spread, Chamfer, Fix, Relieve, Change, Extend, Connect, Pass, Separate, Make, Check, Carry, Cut, Unify, Dualize, Reduce, Expand	43
Prevent	Avoid from some state		0
Purpose	A Task with issues to address		0
SetValue	A Task with value to set		0

Here is a description of classes (Section 3.1) and properties (Section 3.2) of the TCO.

3.1. Class definitions

TCO contains task and non-task classes, such as the Location and the Issue. The overall classes are shown in Fig. 3.

3.1.1. Task class

Table 1 depicts subclasses of the Task Class. There are several subclasses in the Apply class which indicates applying something to a location. What to apply composes the name of each subclass such as the ApplyStructure. So is the Retain class. The Operate class represents material processing such as the OperateHemming. To represent tasks of changing states of some objects, the Modify class is specialized into Connect, Match, Linearize, and so on. It is partially blinded due to the company's security policy. The main discrepancy between the Operate and others is that it involves specialized skills or equipment. The Prevent class aims to express a task that prevents something, usually has some issues to address as illustrated in Fig. 1.

The Purpose and the SetValue classes are defined with necessary conditions. For example, some Task instances that contain a value that needs to be set are instances of the SetValue class. A task with an issue to solve is considered as the Purpose in itself.

3.1.2. Non-task class

There are several non-task classes: Location, Issue, Content, Document, Type, Comparison, and the Negation.

The Location class indicates material components for performing a specific task. Table 2 is a summary of them. Since the domain of this study is limited in scope to a door system of a vehicle, the Location and the VehicleType class reflect it. The Location class contains components of the door and other parts of the vehicle, and the VehicleType contains models and types of vehicle. The Location class is partially inherited from VehiclePart Ontology,² which represents some components of a vehicle.

The Issue class stands for some problems that each task wants to solve or alleviate. It contains subclasses such as Bending, Leakage, and Pollution. The Document class literally represents a document that is the source of a task and the Content represents the content such as the Image and the Table in the document. The DoorType and the VehicleType in the Type class are specialized to the target domain, the automotive closure system. They have several subclasses, such as the Sedan or SlidingDoor. The Comparison class aims to describe the task with baseline, like as "Cut a rebar 4 mm smaller than next one". There are four subclasses of the Comparison: HigherThan, LowerThan, ShorterThan, and LongerThan. Lastly, the Negation class is designed to check constraint violations in knowledge reasoning. It contains conditions that are disjoint from all other classes. Section 4.4 describes the class in more detail.

3.2. Property definition

Table 3 summarizes all object properties in TCO. It is mainly composed of the relations between the task and the non-task classes, between the task classes, and between the non-task classes. In addition, *hasValue* is a data property to describe specific value of a task. For example, triple (Cut, *hasValue*, '4 mm') is extracted from the natural language sentence "Cut a rebar into 4 mm".

Relation between task and non-task indicates a role of task argument. For example in Fig. 1, (EnlargeValue, *hasTargetLocation*, Section) means to enlarge some value in a section. The role of Section for EnlargeValue is the target location in this case. Similarly, the Prevent task on the Corner location may be performed for some issues, such as alleviating deflection and described on a specific document (e.g. RSS-AM-xxxx). The *hasIssue* and the *hasDocument* is for (Prevent, *hasIssue*, Deflection) and (Prevent, *hasDocument*, RSS-AMs-xxxx), respectively.

hasLocation expresses a location that is not the direct target but is related to the task. For example, if a piercing task is performed on a hole in a front door, the *hasTargetLocation* relation between them is extracted. Since the task is about the front door, it can also be expressed as (OperatePiercing, *hasLocation*, FrontDoor). *hasIntervalLocation* expresses an interval between the locations related to a task.

² <https://spec.edmcouncil.org/auto/ontology/VC/VehicleParts/>

Table 2
Subclasses of the location class.

SubClass (level 1)	SubClasses (level 2)	SubClasses (level 3)
Element	Department	MatchingSection, SealingSection,
	Hole	
	Section	
	InterSpace	
MaterialEntity	Point	
	SubComponent	Bracket, Bolt, Fastner, Nut, Screw, Connector, Flange, Clip, Logo, Pad, SealerLine, Hook
	VehiclePart	Door, Panel, w/strip, Trunk, Checker, Latch, Pilar, Lamp, DoorLock, DoorBelt
Assembly		DoorHandle, Glass, DoorHandle, Bumper

Table 3
An overall summary of object properties in TCO.

	Property	Domain Class	Range class	Inverse Property
Task argument	<i>hasTargetLocation</i>	Task	Location	<i>isTargetLocationOf</i>
	<i>hasLocation</i>	Task	Location	<i>isLocationOf</i>
	<i>hasIntervallLocations</i>	Task	Location	
	<i>hasIssue</i>	Task	Issue	<i>isIssueOf</i>
	<i>hasVehicleType</i>	Task	VehicleType	<i>isVehicleTypeOf</i>
	<i>hasDocument</i>	Task	Document	<i>isDocumentOf</i>
	<i>hasComparison</i>	Task	Comparison	
Relation between Task instances	<i>hasSubTask</i>	Task	Task	<i>isSubTaskOf</i>
	<i>hasAlternativeTask</i>	Task	Task	
Relation between NonTask instances	<i>partOf</i>	Location	Location	<i>hasPart</i>
	<i>meetSpatially</i>	Location	Location	
	<i>isLongerThan</i>	Location	Location	
	<i>isShorterThan</i>	Location	Location	
	<i>isLowerThan</i>	Location	Location	
	<i>isHigherThan</i>	Location	Location	
	<i>hasComparedObject</i>	Comparison	Location	
	<i>hasReferenceObject</i>	Comparison	Location	
	<i>hasDoorType</i>	Door	DoorType	<i>isDoorTypeOf</i>
	<i>containsImage</i>	Document	Image	
	<i>containsTable</i>	Document	Table	
	<i>containsTask</i>	Table	Task	
	<i>associatedWith</i>	Issue	Location	

The textual sentence “to set the interval between A and B” is converted to (ApplyInterval, *hasIntervallLocation*, A) and (ApplyInterval, *hasIntervallLocation*, B). *hasComparison* indicates a task that requires a comparison. For instance, “setting the length of C to be B longer than A” is represented by (ApplyLength, *hasComparison*, LongerThan).

Relation between tasks Tasks in the same document may relate to each other. *hasSubTask* represents that one is an assistance task of the other and *hasAlternativeTask* is to represent substitutive tasks. Suppose a water bowl is overflowing a user may drain the water out of the bowl or move the water into a larger bowl. It is derived into (DrawOff, *hasAlternativeTask*, Move) and (Move, *hasAlternativeTask*, DrawOff).

Relation between non-tasks There is a semantic connection between the non-task classes, such as the Location and the Issue. A door may consist of an inner panel, and the panel may contact with a side panel. There are (InnerPanel, *partOf*, Door) and (InnerPanel, *meetSpatially*, SidePanel) in this situation.

There are several properties associated to Comparison: *isLongerThan*, *isShorterThan*, *isLowerThan*, and *isHigherThan*. They express that a location has a different state relative to another location in order to perform a particular task. From the phrase “setting the length of C to be Bmm longer than A”, we extract factual relations like (ApplyLength, *hasComparison*, LongerThan), (LongerThan, *hasComparedObject*, C) and (LongerThan, *hasReferenceObject*, A). For a concrete description, the relation between A and C may be (C, *isShorterThan*, A) or (A, *isLongerThan*, C). An additional data property, *hasDifferenceValue*, can be applied as in (LongerThan, *hasDifferenceValue*, Bmm).

hasDoorType indicates that a door has a door type, such as sliding door or swing door. *containsImage* and *containsTable* imply whether a document contains images or tables. *containsTask* literally ensures the

table in document describes a task as with *containsValue*, as shown in Fig. 1(c). *associatedWith* is for an implicit relation between issue and location that are roles in the same task.

4. Ontology-based information extraction framework

For a given manufacturing document $D = \{(X, I, T)\}$ consisting of $X = \{x_i\}_{i=1}^{n_x}$ with n_x sentences, $I = \{m_i\}_{i=1}^{n_m}$ with n_m images, and $T = \{t_i\}_{i=1}^{n_t}$ with n_t tables, the goal is to construct task-centric KG G according to TCO. Accordingly, G is an ensemble of triple (e_1, r, e_2) consisting of a single relation and two instances of domain and range classes (c_{subj} and c_{obj}) corresponding to property schema of r .

Fig. 4 shows an overall architecture of our framework, OmEGa. It mainly decomposes into four parts, (1) Document Preprocessing with multimodal and layout, (2) Multimodal Information Extraction, (3) Modality Linking and (4) Knowledge Reasoning. To handle multimodal data, information from textual and non-textual components are separately managed. For effective IE, T and I are textualized into X_T and X_I in the first part. In the second one, the system recognizes instances in X , X_T and X_I and then extract relations between the instances of the same modality. $\circ \mid \bowtie \mid \sqcup$ The third part aims to integrate all the knowledge components present in the document to alleviate the structural challenges. The last process expands and validates the inadequate extraction based on TCO.

4.1. Document preprocessing

To fully understand D , two preprocessing processes are conducted: Multimodal and Layout.

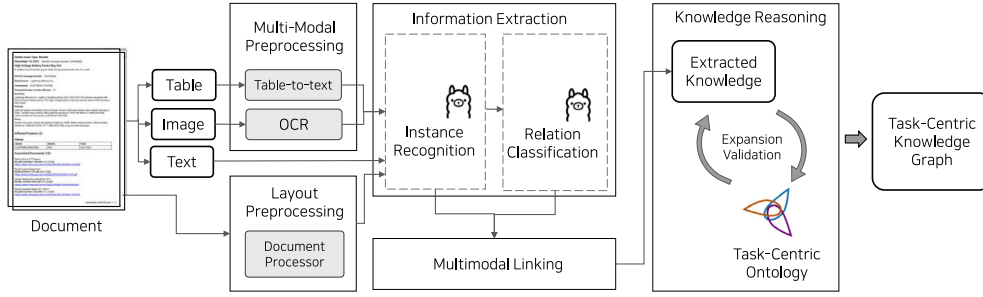


Fig. 4. Overall architecture of OmEGa.

4.1.1. Multi-modal preprocessing

In manufacturing documents, tables T and images I commonly serve as evidence or supplementary explanations for corresponding text X_T and X_I . Unlike typical multimodal documents, I often contains characters or numbers rather than photographs of objects which are open-domain. An Optical Character Recognition (OCR) model aims to capture words from I . In this paper, an off-the-shelf OCR system³ is utilized.

$$X_I = OCR(I) \quad (1)$$

$$X_T = \text{linearize}(T) \quad (2)$$

Text from tables may also be generated. Due to the urgent nature of these documents, T often has irregular rows and columns, making it difficult to apply standard table-to-text models. To address this, tables are linearized into a $\{\text{column} : \text{component}\}$ format. Any table-to-text system may allow based on the specific characteristics of the data.

4.1.2. Layout preprocessing

With the given document D , 2D positional information of all the components is derived first. This is represented in a bounding box, which contains a [left index, right index, top index, bottom index] of each word. We employ PyMuPDF⁴ as a document processor to extract bounding boxes of all words in D .

The spatial information acts as a cue for extraction by grasping spatial information of each component in D . It is used to recognize whether the words in D are instances or not. This spatial clue is the material of layout embedding in Section 4.2.1. By integrating spatial embedding with existing embeddings such as 1D position or word, the model enhances the capability for document understanding.

4.2. Information extraction

Given text $X_D = \{X, X_T, X_I\}$ from a input document D , this module aims to extract instances and their relations based on TCO. Not all classes are targeted to extract instances; for example, document IDs (Document) and tables and images (Content) are excluded from the extraction process since they exist to model a structure for D . Nor are instances of the Negation class, which is used for knowledge reasoning, and the Comparison class, which expresses comparisons. We focus on recognizing instances of the following classes: Task, Locations, Issue, and the Type (VehicleType and the DoorTypes). In addition, We excluded the primitive classes such as the Purpose and the SetValue since they naturally expand in knowledge reasoning.

Likewise, only relationships that cannot be inferred through rules or understood without parsing the text are classified. Additionally, any relation that occurs less than 3 times in the document, such as *isLongerThan* and *isShorterThan*, is omitted. The actual relations to

Table 4

An exmple of data augmentation for instance recognition. The translated version of example is 'Welding for broken door'.

Process	Example sentence
original	DR 내측 터짐 위한 용접
masked	[door] 터짐 위한 용접
replaced	문 터짐 위한 용접
	DOOR 터짐 위한 용접
	도아 터짐 위한 용접

extract are task arguments (*hasTargetLocation*, *hasIssue*, and *hasVehicleType*, *hasIntervalLocations*), relations between the Location instances (*partOf* and *meetsSpatially*), and relations between the Task instances (*hasAlternativeTask*, and *hasSubTask*).

4.2.1. Instance recognition

This module aims to extract instances predefined in TCO in text X_D . As illustrated in Fig. 5(a), we employ a LS-Llama model [15], leveraging Low-Rank Adaptation (LoRA) [31]. It allows parameter-efficient fine-tuning since only a linear layer is trained while a LLM (LlaMA) is frozen. This advocates for efficient training of LLM which is suitable for real-world industries with restricted costs for training.

To mitigate the contextual and structural challenges when training the backbone model, two methods are further applied: Layout embedding and Data augmentation. Layout embedding layer aims to convert spatial position information into layout embedding. Layout embedding layer aims to convert spatial position information into layout embedding. As in LayoutLMv3 [26], the embedding that allows for training 2D position information is trained. Specifically, spatial clues such as bounding boxes are fed in the layout embedding layer so the model may be aware of the structure and composition of the document to improve its ability to understand unstructured multimodal documents. It enables a more precise extraction of target entities since where they are located is a significant clue. This leads to alleviating structural challenges in multimodal manufacturing document understanding. X with position feature is then to-kenized to feed into the module.

In addition, the train split of our target is augmented to improve the model's ability to recognize and interpret specialized terminology. Table 4 introduces an example of the augmentation. We first mask all mentions, such as 'DR' in X with corresponding instance types, such as 'door'. Then, randomly replace them with other mentions that appear in other training examples. This may expose the model to various mentions for each instance, enhance its proficiency with domain-specific terminology, and consequently improve its capability to understand co-references.

$$Loss_{IR} = -\frac{1}{M} \sum_{i=1}^M \sum_{j=1}^C y_{ij} \log(\hat{y}_{ij}) \quad (3)$$

A classical cross-entropy loss, $Loss_{IR}$ in Eq. (3), is utilized to train the Instance Recognition module. M indicates the total number of

³ https://github.com/yunwoong7/korean_ocr_using_pororo

⁴ <https://pymupdf.readthedocs.io/>

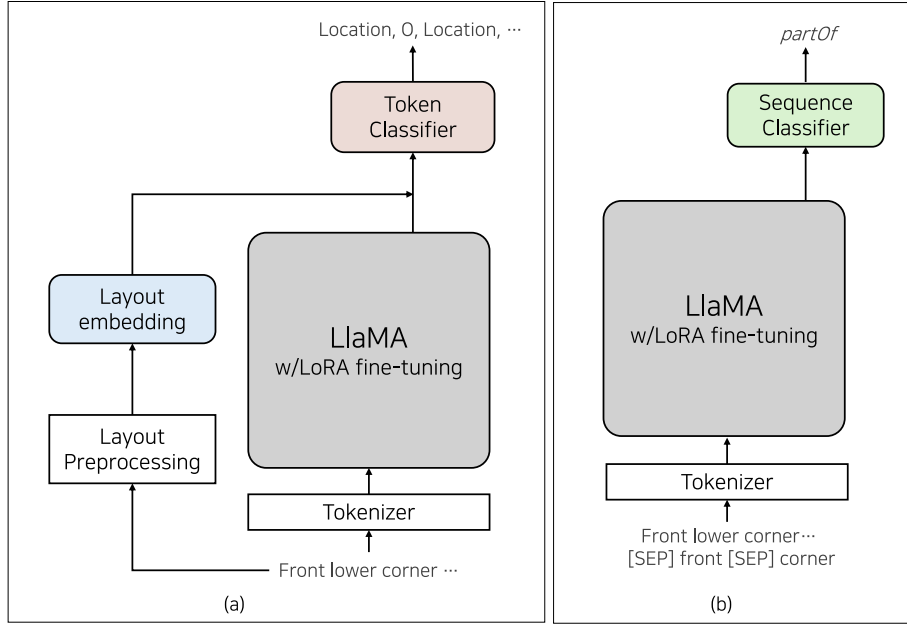


Fig. 5. Architectures of Information extraction module. (a) Instance recognition, and (b) Relation classification.

tokens in the dataset. Each token corresponds to a word unit that the model attempts to classify into an entity category. C is the total number of entity classes, including all named entity types for tokens that do not belong to any entity. y_{ij} is an one-hot encoded binary indicator that is 1 if the i th token belongs to the j th entity class and 0 otherwise. Lastly, $\hat{y}_{ij}$ predicted probability that the i th token belongs to the j th entity class, output by the model.

4.2.2. Relation classification

Similar to the instance recognition, we leverage LS-LLaMA [15] to classify relations in given instance pairs depicted in Fig. 5 (b). Each instance pair contains subject and object instances with their mention, instance type, and start and end indices, respectively.

$$Loss_{RC} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^R y_{ij} \log(\hat{y}_{ij}) \quad (4)$$

The notations of $Loss_{RC}$ in Eq. (4) are as follows. N is the total number of entity pairs in the dataset. Each entity pair corresponds to a potential relation that the model attempts to classify. K implies the total number of relation classes, including both the specific relation types and a *no relation* for pairs with no meaningful relationship. y_{ik} is a one-hot encoded binary indicator, which means 1 if the i th entity pair belongs to the k th relation class and 0 otherwise. $\hat{y}_{ik}$ is the predicted probability that the i th entity pair belongs to the k th relation class, output by the model.

4.3. Modality linking

Although non-textual information, output from X_I and X_T as derived in Eqs. (1) and (2) and textual information, output from X are acquired in the same manner, the modality difference in source components distinguishes them. Nevertheless, ensuring unified knowledge from each document is crucial. Non-textual information may link with textual information. Since modality linking fuses information from different modalities, structural challenges are mitigated, and a resulting knowledge graph is enriched.

To do this, instances of non-textual output that are identical to those in the textual instances are linked using the *rdf:sameAs* relation. As depicted in Fig. 1, orange boxes located in both text and image imply the same instances. Modality linking affords to make *rdf:sameAs* between them. The same goes for gray boxes in text and tables.

4.4. Knowledge reasoning

The triples extracted in Sections 4.2 and 4.3 may be expanded and validated based on schema-based reasoning. Table 5 demonstrates some examples of schemas for knowledge reasoning.

Leveraging TCO schema, OmEGA can complement the original source or the extraction if any information is excluded from the document or extracted incorrectly or missed during the information extraction process. For example, a piercing that drills a hole always has a hole as the *targetLocation* argument, as stated in Table 5 (3). Schemas such as Table 5 (4–5) contribute to detect the implied meanings between instances within a document and link them. Reflective relationships (Table 5 (6)) and inverse relationships (Table 5 (7–9)) are also adopted to the knowledge reasoning. So as some primitive classes like the *Purpose* and the *SetValue*. As illustrated in Fig. 1(c), the *Prevent* instance containing *hasIssue* has the information added that it is of type *Purpose*. Similarly, the *ReduceDepth* instance with *hasValue* can extend the knowledge that it is of type *SetValue*.

Moreover, the reasoning also allows us to correct inconsistencies or inaccuracies in the extracted information by validating based on the schema. For example, if there is (Door, *partOf*, Panel) in the KG, it needs to be corrected in doorSystem domain, which contains door as a top-level component. As noted in Table 5 (1), Door becomes *Negation* instance and a disjoint condition (Table 5 (9)) occurs inconsistency. This alerts a domain expert about incorrect knowledge and necessity to correct. Table 5 (2) indicates that the *MatchedSection* must always have at least two matching targets, resulting in at least two *partOf* relations with two instances. The schema plays a role as a validating indicator in the same manner. Moreover, disjoint conditions such as Table 5 (10) also contribute to validate knowledge.

The invalid knowledge discovered in the knowledge reasoning may be corrected with a human-in-the-loop system as in Fig. 6. It enables elaborate and exhaustive rectification, resulting in the intervention of a domain expert. This is crucial in the manufacturing domain, where precise prediction is essential, as it enables the model to correct inadequate predictions and enhance the overall reliability of the results. Pellet reasoner [32] in Protégé is used.

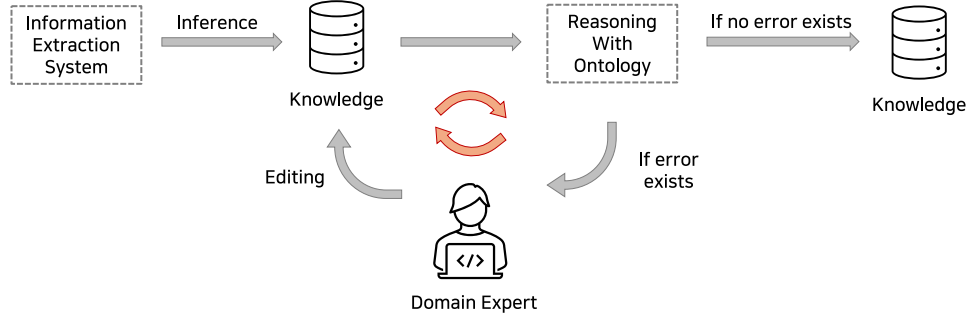
5. Experiments

This section indicates experiments to demonstrate the utility of TCO and the performance of OmEGA.

Table 5

Examples of TCO utilized for knowledge reasoning. Only a subset of the schema is revealed due to the security policies of the source company.

Schema	
(1)	<code>Door(?x), partOf(?x,?y) ->Negation(?x)</code>
(2)	<code>MatchedSection(?x), count(?p, ?n), partOf(?x, ?p), lessThan(?n, 2) ->Negation(?x)</code>
(3)	<code>Piercing(?x) ->hasTargetLocation(?x, Hole)</code>
(4)	<code>hasIssue(?x, ?y), hasTargetLocation(?x, ?z) ->associatedWith(?y, ?z)</code>
(5)	<code>hasTargetLocation(?x, ?y), partOf(?y, ?z) ->hasLocation(?x, ?z)</code>
(6)	<code>hasAlternativeTask(?x, ?y) ->hasAlternativeTask(?y, ?x)</code>
(7)	<code>hasTargetLocation(?x, ?y) ->isTargetLocationOf(?y, ?x)</code>
(8)	<code>partOf(?x, ?y) ->hasPart(?y, ?x)</code>
(9)	Negation disjoint with Comparison, Content, Document, Issue, Location, Task, Type
(10)	FrontDoor disjoint with RearDoor

**Fig. 6.** The flow of Knowledge reasoning with human-in-the-loop manner.**Table 6**

Statistics of datasets in terms of documents and sentences.

Dataset	Document			Sentences		
	train	dev	test	train	dev	test
DoorSystem	326	43	42	1,637	209	214
w/augmentation	3,260	43	42	16,370	209	214
Automobile Recall	394	46	60	2,947	349	464

5.1. Data

We assess OmEGa on two manufacturing documents: DoorSystem Dataset and Automobile Recall Dataset. Tables 6 and 7 present their statistics. Five crowdworkers, who are well-acquainted with the domain, annotate datasets according to TCO. We manually split the train, dev, and test sets to ensure that we had an even distribution for all classes and relations.

5.1.1. DoorSystem dataset

The DoorSystem dataset is from real-world industries, especially for producing and maintaining the door system of vehicles. Fig. 1 is an example of a blinded version of the dataset. The data is a main target to develop TCO and extracting knowledge.

It contains texts, tables, and images in Korean with some English and Chinese characters. Since Korean is an agglutinative language, which can decompose all phrases into stem and end, this data is highly abbreviated by omitting the ends in sentences. For example, '1안. 문 내부의 터짐을 방지하기 위한 용접을 수행한다.' (Case 1. Welding is performed to prevent bursting inside the door.) may be notated '1案: DR 내측 터짐 위한 용접 (C1: Welding for DR inner rupture)'. It makes extraction extremely challenging because Josa (조사), which determines part-of-speech (PoS) of each word in the sentence, is totally omitted. Moreover, the missing end confuses the distinguishing between nouns and verb stems. In addition, this dataset contains insufficient amounts, as shown in Table 6. We can infer this dataset is significantly more difficult than Automobile Recall Dataset.

5.1.2. Automobile recall dataset

It is unable to disclose a full version of the DoorSystem Dataset since it is used in practice and contains numerous valuable techniques of specific enterprises. Instead, Automobile Recall dataset,⁵ as illustrated in Fig. 7, is also used. The dataset contains recall information related to automobiles, specifically in English. It also includes descriptions of actions taken by manufacturers when a product is defective or does not comply with safety standards. In this study, 500 examples are randomly selected to ensure a representative sample.

Unlike the DoorSystem dataset, this is an open dataset that covers a broad range of defects in the automobile industry. The data is well-formatted, with no abbreviations, allowing for favorable analysis and interpretation. It further enables us to confirm the generalizability of TCO with overall manufacturing domains.

5.2. Baselines

A baseline consists of two main parts. The first is a set of models that evaluate the recognition of instances, and the second is to classify relations between the instances.

5.2.1. Instance recognition

Multilingual PLMs Since the datasets consist of multilingual text, particularly in English and other languages, multilingual BERT [33] and multilingual RoBERTa [34] with additional linear layers are concluded as comparative models. This allows us to evaluate the performance across various linguistic contexts. The base version of them is employed due to considering the low-resource setting.

LS-Llama It is natural to adopt vanilla LS-Llama [15] as a baseline owing to the backbone of OmEGa, which follows LS-Llama. The comparisons based on the Llama-based LLMs may be discussed in Section 6.1.1.

⁵ <https://www.kaggle.com/datasets/michaelbryantds/automobile-recalls-dataset>

Table 7
Statistics of datasets in terms of instances and relations.

		DoorSystem			Automobile		
		train	dev	test	train	dev	test
Instances	Task	35,500	370	338	1,107	145	180
	Issues	9,955	95	86	1,161	143	194
	Location	60,073	612	605	855	134	138
	Others	1,642	26	28	0	0	0
	Total	106,770	1,103	1,057	3,123	422	512
Relations	between Tasks	4,338	459	430	0	0	0
	between NonTasks	2,105	202	257	0	0	0
	between Task and NonTask	7,234	793	764	1,289	189	223
	Total	13,677	1,454	5,361	1,289	189	223

Safety Issue Type: Recalls
December 16 2022 NHTSA Campaign Number: 22V934000
High Voltage Battery Packs May Fail
A sudden loss of traction power while driving increases the risk of a crash.

NHTSA Campaign Number 22V934000
Manufacturer Lightning eMotors Inc.
Components ELECTRICAL SYSTEM
Potential Number of Units Affected 27
Summary
Lightning eMotors Inc. (LeM) is recalling certain 2021-2022 FE4-129 vehicles equipped with Romeo Power battery packs. The high voltage battery may lose traction power while driving or fail to start.
Remedy
LeM will replace the battery, free of charge. Owner notification letters were mailed February 6, 2023. Owners may contact LeM customer service at 1-970-744-4926 or 1-800-223-0740. LeM's number for this recall is LeM Recall 2022-002.
Notes
Owners may also contact the National Highway Traffic Safety Administration Vehicle Safety Hotline at 1-888-327-4236 (TTY 1-800-424-9153), or go to www.nhtsa.gov.

Affected Products (2)

MAKE	MODEL	YEAR
LIGHTNING EMOTORS	FE4	2021-2022

Associated Documents (10)

Defect Notice 573 Report
RCLRP22V934-1166.PDF 214.727KB
<https://static.nhtsa.gov/od/rlc/2022/RCLRP22V934-1166.PDF>

Recall Acknowledgement
RCAK22V934-7137.pdf 645.519KB
<https://static.nhtsa.gov/od/rlc/2022/RCAK22V934-7137.pdf>

Owner Notification Letter(Part 577)
ROCNL-22V934-0432.pdf 273.255KB
<https://static.nhtsa.gov/od/rlc/2022/ROCNL-22V934-0432.pdf>

Recall Quarterly Report #1, 2023-1
RCLQRT-22V934-1333.PDF 211.136KB
<https://static.nhtsa.gov/od/rlc/2022/RCLQRT-22V934-1333.PDF>

Generated at NHTSA.gov 1 / 2

Part 573 Safety Recall Report 22V-934

OMB Control No.: 2127-0004

Manufacturer Name: Lightning eMotors Inc.
Submission Date: DEC 16, 2022
NHTSA Recall No.: 22V-934
Manufacturer Recall No.: LeM Recall 2022-002

Manufacturer Information:
Manufacturer Name: Lightning eMotors Inc.
Address: 815 14th Street SW
Suite A100 Loveland CO 80537
Company phone: (970)744-7241

Population:
Number of potentially involved: 27
Estimated percentage with defect: 100 %

Vehicle Information:
Vehicle 1: 2021-2022 Lightning eMotors FE4-129
Vehicle Type: BUSES, MEDIUM & HEAVY VEHICLES
Body Style: OTHER
Power Train: HYBRID ELECTRIC
Descriptive Information: All Lightning eMotors FE4-129 vehicles equipped with Romeo Power battery packs are included in this recall. FE4-129 vehicles equipped with eMatrix battery packs are not part of the recall.

Lightning eMotors/FE4-129 Qty 27
Production Dates: JAN 01, 2021 - OCT 21, 2022
VIN Range 1: Begin: NR End: NR ☐ Not sequential

Description of Defect:
Description of the Defect: Due to multiple software and hardware discrepancies internal to the Romeo Power battery packs installed in certain Lightning eMotors (LeM) FE4-129 series vehicles, the vehicle may fail to operate in cold temperatures, fail to start when the key is placed on, or may lose traction power while driving.
FMVSS 1: NR
FMVSS 2: NR
Description of the Safety Risk: A loss of traction power while driving may increase the risk of a crash.
Description of the Cause: Lightning eMotors has identified that there are multiple faults internal to the Romeo Power 30.4kWh Battery Packs installed on the subject vehicles. These include:
1) Flood Sensor fault
2) HVIL fault
3) Cell Imbalance fault

The information contained in this report was submitted pursuant to 49 CFR 5573

Fig. 7. An example of Automobile Recall dataset.

5.2.2. Relation classification

Identical models are adopted as the baselines for relation classification. In addition, since the TCO inherits the Event Ontology, there is a high similarity between the Event and the Task. The relation classification between Task instances indicates Task RE. Moreover, relations between nonTask instance and Task instance resemble event arguments and their roles, in other words, task argument extraction. This leads us to complement the baselines as follows. It may extend to other dataset due to the minority and difficulty of the DoorSystem dataset.

Task Argument Extraction Models BART-GEN [35] is a seq-to-seq method that conditionally generates arguments with a given event template and document context. PAIE [36] selects argument spans leveraging corresponding event prompts. We utilize task prompts based on a definition of each task. DEGREE [37] is a model specialized for low-resource settings such as insufficient datasets. Since TabEAE [38] is a current State-of-The-Art model of EAE, it is natural to be adopted for additional comparison. TabEAE extracts arguments considering all events that exist in a document.

Task RE Models Wang et al. [39] propose relations between events such as causal or subevent. We compare to extract *hasSubTask* and *hasAlternativeTask* in a given document.

5.3. Evaluation metrics

OmEGa's performance is evaluated with precision, recall, and f1 score.

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}} \quad (5)$$

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (6)$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

Precision is the ratio of true positive predictions to the total number of positive predictions made by the model. It measures how accurately the model identifies the relevant instances, with a high precision indicating that once the model makes predictions, it is highly likely to be correct. Recall is the ratio of true positive predictions to the total number of actual positive instances in the dataset. It reflects the model's ability to capture all relevant instances, with a high recall indicating that the model produces few false negatives.

F1 score is the harmonic mean of precision and recall, indicating a balance in the trade-off between these two measures. It determines the overall performance of the model in terms of both Precision and Recall.

5.4. Implementation details

As mentioned in Sections 4.2.1 and 4.2.2, Llama-based LLMs are adopted as a backbone of OmEGa. Concretely, we employ Llama2 7B, Ko-Llama2 7B, and 42dot in OmEGa (42dot), OmEGa (ko-Llama2), OmEGa (Llama2), respectively. They contain parameters less than 8B among other Llama families. OmEGa is able to handle multilingual documents due to the backbone models, which were trained on a large corpus of multilingual datasets.

Table 8

Overall performance of instance recognition. Best score is marked with an underline and bolded score is for secondary best.

Model	DoorSystem			Automobile Recall		
	Precision	Recall	F1 score	Precision	Recall	F1 score
mBERT-NER	69.99	66.76	68.34	96.00	91.13	93.50
mRoBERTa-NER	47.48	29.54	36.42	93.30	88.37	90.77
LS-Llama	47.64	39.80	43.37	96.78	87.08	91.67
OmEGa (42dot)	74.01	69.17	71.51	99.18	95.87	97.50
OmEGa (ko-Llama2)	71.91	67.10	69.42	96.01	91.03	93.45
OmEGa (Llama2)	50.83	44.21	47.28	93.44	90.83	92.12

Table 9

Overall performance of relation classification. Best score is marked with an underline and bolded score is for secondary best.

Model	DoorSystem			Automobile Recall		
	Precision	Recall	F1 score	Precision	Recall	F1 score
mBERT-RE	80.96	87.72	84.21	72.22	96.29	82.53
mRoBERTa-RE	77.05	<u>84.97</u>	80.81	37.59	<u>56.08</u>	45.01
OmEGa (42dot)	84.72	84.13	84.43	67.11	79.89	72.94
OmEGa (ko-Llama2)	84.51	87.31	85.88	83.01	90.47	86.58
OmEGa (Llama2)	80.06	76.14	78.05	88.33	84.12	86.17

Table 10

Overall performance of task argument extraction. Note that this score is a combination of instance recognition and relation classification. This is because the objective of argument extraction identifies instances and their role simultaneously.

Model	DoorSystem		
	Precision	Recall	F1 score
BART-Gen [35]	33.90	22.30	26.90
PAIE [36]	1.56	1.80	1.67
DEGREE [37]	43.04	45.95	44.44
TabEAE [38]	47.1	25.2	32.8
OmEGa (ko-Llama2)	90.13	56.21	66.86

Llama2 7B is trained on 2 trillions of multilingual tokens such as Korean and English. It is able to handle context length 4096 and contains 32,000 of vocabulary size. Ko-Llama2 7B is an advanced version of Llama2 7B with an expanded vocabulary size, 46,336, and the inclusion of a Korean corpus in its further pretraining. 42dot is trained on 1.4T tokens based on Korean and English datasets such as Wikipedia,⁶ RedPajama,⁷ and so on. It also manages a context length of 4096.

For instance recognition, we trained our model for 10 epochs with a learning rate of $2e-5$ and a batch size of 1. The total training time for this task is approximately 40 min. For relation classification, the model is trained for 2 epochs with a learning rate of $3e-5$ and a batch size of 8, resulting in a total training time of around 3 h.

All experiments are conducted on a hardware setup featuring a single NVIDIA A100 GPU with 40 GB of memory. This setup provides sufficient computational power to handle large-scale training tasks efficiently.

5.5. Main result

Tables 8 and 9 show summarized performances of OmEGa compared to other baselines. It archives the best or secondary best performance for all metrics of all datasets.

5.5.1. Performance of information extraction

As shown in Tables 8 and 9, OmEGa outperforms existing models such as LS-Llama and multilingual PLM-based supervised fine-tuning. In particular, it achieves a score 2–3 times higher than the baseline on the

Table 11

Overall performance of task RE.

Model	DoorSystem		
	Precision	Recall	F1 score
Wang et al. [39]	49.32	50.00	49.66
OmEGa (ko-Llama2)	81.00	86.48	83.65

Table 12

Performance according to the knowledge reasoning.

	Precision	Recall	F1 score
w/o knowledge reasoning	81.00	91.05	85.73
w/knowledge reasoning	85.46	92.62	88.90

instance recognition task on the DoorSystem dataset than the existing models on all metrics. It also demonstrates superior performance in the relation classification task on the same dataset. OmEGa's advantage is especially evident in its higher precision scores, emphasizing the nature of the manufacturing domain that is precise prediction is crucial.

The Automobile Recall dataset, while still a domain-specific dataset, covers more general terms and fields compared to the DoorSystem dataset. It is natural that other baselines also perform reasonably well on this dataset. Nonetheless, OmEGa consistently performs the best or the second-best model, indicating the reliability of its robustness.

Furthermore, OmEGa outperforms traditional event argument extraction models, as in Table 10. It significantly surpasses models like DEGREE [37], which is designed for low-resource settings, and BART-GEN [35], and PAIE [36], which utilize event schemas. Notably, OmEGa achieves precision above 90, which is particularly significant in domains where accurate predictions are more critical than recall. Table 11 indicates that OmEGa surpasses traditional Event Relation Extraction models, even those tailored to consider the specific nature of events. This confirms the superiority of our parameter-efficient model in a variety of challenging scenarios. After Information Extraction is performed, the number of triples generated in the DoorSystem dataset is 1929 as shown in Table 13.

5.5.2. Performance of knowledge reasoning

Knowledge reasoning assists the incomplete extraction result by leveraging the task-centric schema in a human-in-the-loop manner. Table 12 demonstrates its remarkable performance in constructing task-centric KG. It improves the triple-level performance from 85.73 to 88.90 in the aspect of the F1 score. It is noticeable that the precision increases 4.46 before reasoning.

⁶ <https://www.wikipedia.org/>

⁷ <https://github.com/togethercomputer/RedPajama-Data>

Table 13
Number of triples with their relations in the point of the knowledge reasoning.

		Object property	Data property	Type information	Total relations
Inferred	w/o reasoning	1,119	293	517	1,929
	w/reasoning	3,967	493	1,648	6,108
Golden	w/o reasoning	1,059	324	504	1,887
	w/reasoning	3,487	524	1,591	5,602

Table 13 indicates the knowledge expansion result with a number of triples. They are expanded into 5 times in view of all property types. Type information rises in particular, which contains several constraints with the Negation class.

6. Discussion and analysis

We demonstrate the robustness of the proposed method by conducting an ablation study and further analysis. Since the DoorSystem dataset is more challenging than the other, we do perform a comprehensive analysis with this dataset. Also, the use case demonstrates the utility of TCO and task-centric KG.

6.1. Ablation study

Ablation study is accomplished with three settings: in terms of LLM (Section 6.1.1), layout embedding and data augmentation (Section 6.1.2).

6.1.1. Effectiveness of language model

We implement OmEGa with various Llama-based LLM : 42dot(1.3B), Ko-Llama2(7b) and Llama2(7b). This is because of the language of the source document, which contains Korean and English, and restricted resources for training (up to 7b). As shown in Tables 8 and 9, our OmEGa shows robust and substantial performance in all datasets and tasks.

Noticeably, there is no correlation between the size of LLM parameters and performance in the DoorSystem dataset. This indicates that this dataset is so challenging and exclusive with other benchmarks that it has less influence on the general performance of LLM. Considering that the dataset is commonly found in the real world, we can infer a novelty in OmEGa's ability to extract accurate information from industry.

6.1.2. Effectiveness of layout embedding and data augmentation

Table 14 shows the informativeness of layout embedding in document understanding. A combination of data augmentation and layout embedding induces better performance in instance recognition. We establish that the data augmentation is intimate with unaccustomed terminology. Noticeably, OmEGa only with augmented data leads to higher precision. This indicates that data augmentation facilitates LLM's adaptation to unfamiliar contexts, mitigating the contextual challenges.

Table 14 also implies that the spatial information of each component in the document may offer supplementary clues to identify instances. An absence of layout embedding results in lower recall. It establishes that OmEGa accommodates the structural challenges that multimodal manufacturing documents usually contain. Moreover, using the Layout Embedding gives an approximate performance improvement as ten times of augmenting the training data. Considering that an increased amount of training data requires a significant training cost, we can infer that the embedding also effectively contributes to compensating for the contextual challenges of manufacturing documents.

6.2. In-depth analysis

In this section, further analysis of the aspect of task and task argument is conducted. OmEGa with 42dot is employed since it records the highest score.

Table 14
Ablation result of instance recognition.

	Precision	Recall	F1 score
OmEGa (42dot)	74.01	69.17	71.51
w/o data aug.	75.72	64.26	69.52
w/o layout	74.40	68.8	71.48
w/o data aug. & layout	68.84	65.84	67.3

6.2.1. Analysis of instance recognition

Fig. 8 and Table 15 depict confusion matrix and concrete performance, respectively.

Performance difference between class types Fig. 8(a) shows the confusion matrix for the Task, Type, Issue, and Location classes. It can be seen that the Task and the Location classes have the most true positives and confuse each other in prediction. This is mainly due to the fact that some locations have mention Task mentions in their names, such as the HemmingSection and the SealingSection. Otherwise, we can see that the model classifies well overall. However, for the Type class, the performance is relatively low due to the small data frequency.

The most challenging Task instances Fig. 8(b) and Table 15 show that the best performance is for the Prevent, and the least is for the Operate. We can infer that there is the lowest variety of mentions in terms of word count, making it relatively easy to train the model. While the Retain, which has a wide range of mention lengths, performs relatively poorly, the Apply performs better despite the same tendency. This indicates that a sufficient amount of data alleviates the difficulty of diverse mention sizes. Moreover, mentions in the Operate are highly likely to be subequal terminologies of the manufacturing domain. In other words, they are words that LM is unlikely to have encountered in the pre-train phase. As a result, its performance is lower than other classes.

6.2.2. Analysis of relation classification

Table 16 is a summarization of relation classification performance. It also contains relation types and their counts in test set.

Performance difference between relation types For relation classification, there is some performance variation in each relation. The model performs proficiently for *hasIssue* and *hasDoorType*, where it is relatively easy to infer relationships from words alone, and for *hasSubTask* and *partOf*, where we have enough data. However, for *hasAlternativeTask* and *meetsSpatially*, the model made no predictions at all. We propose a way to improve this. First, in *hasAlternativeTask*, a formal word such as 'or' is used to mean that two tasks are alternatives to each other. This relation may be automatically extracted when a specific word appears. Also, considering that *meetsSpatially* occurs in a specific Location, such as the MatchingSection, It may be supplemented by defining and applying rules during the extraction process.

Correlation of performance and data amount As represented in Table 16, overall performance is proportional to the number of relationships. Performance is outstanding for *hasIssue* and *no_relation* with a large number of data and poor for *hasAlternativeTask* and *meetsSpatially* with sparse data counts. *hasDoorType* has about the same amount of data as *hasVehicleType*, but shows completely different trends in precision and recall. For *hasVehicleType*, the model confuses 'which vehicle

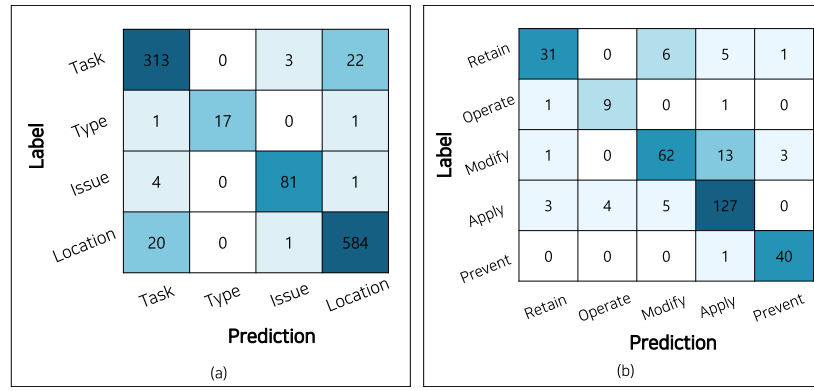


Fig. 8. Confusion matrix of instance recognition. (a) Overall instance recognition (b) Task instances recognition.

Table 15

Instance Recognition Performance of Task class with corresponding counts of mention and instance.

Subclass	Precision	Recall	F1 score	# Instances	# Mentions	Minimum word count of mention	Maximum word count of mention
Retain	86.1	72.1	78.5	401	181	1	10
Prevent	90.9	97.5	94.2	440	53	1	2
Apply	86.3	91.4	88.8	1487	619	1	10
Modify	84.9	78.5	81.6	631	282	1	6
Operate	69.2	81.8	75.0	284	106	1	5

Table 16

Relation classification performance with relation type and count in test set.

Type	Name	Count	Precision	Recall	F1 score
Between Task & NonTask	<i>hasIssue</i>	427	94.8	97.7	96.2
	<i>hasTargetLocation</i>	239	70.2	83.7	76.3
	<i>hasIntervalLocation</i>	36	58.8	27.8	37.7
	<i>hasVehicleType</i>	17	60.0	88.23	71.42
Between Tasks	<i>hasSubTask</i>	427	86.4	84.5	85.4
	<i>hasAlternativeTask</i>	2	0	0	0
Between NonTasks	<i>partOf</i>	238	81.6	87.8	84.6
	<i>meetSpatially</i>	3	0	0	0
	<i>hasDoorType</i>	16	100.0	62.5	76.9
All types	<i>no_relation</i>	3626	95.5	94.3	94.9

type corresponds to which task' compared with *hasDoorType*, confusing 'is a door type defined'. In addition, *hasIntervalLocation* has more data than the previous two, but it is confused with *hasTargetLocation*, missing all the locations that compose the interval.

6.3. Use case: Question answering on task-centric KG

We demonstrate a question-answering (QA) system based on task-centric KG as a use case to reveal its potential applications since questioning and answering based on the manufacturing domain is commonly conducted. We assume that there is no sufficient QA data pair to train any KGQA system, considering real-world situations. So competency questions and their answers are utilized as QA data to build the KGQA system.

In KGQA, KG is a source knowledge for answering questions. The system answers a question by exploring the KG to find a path from entities in the question to an answer entity. This exploration path, which is a reasoning path, acts as a rationale for an answer. To do this, the system first recognizes all the TCO instances in a given natural language question q . The instances play a key role in linking a question and the KG. Then a formal query Q to be executed on the task-centric KG is generated in a rule-based manner. Q aims to find specific documents, which contain key instances from q , and extract the instances corresponding to answers in the documents. The result of executing Q comes to the answer to q . Unlike typical QA systems answering on

unstructured and messy original documents, the reasoning path of the KGQA system acts as a rationale enhancing the user's comprehension.

This system may serve as an application to enhance the interaction between domain experts. Fig. 9 an example of its demonstration with the DoorSystem dataset shown in Fig. 1. The QA system is adaptable to any language version. For convenience, all contents in Fig. 9 are translated into English. The corresponding document and subgraph of a given question are captured along with clue instances such as word 'Corner' and the pink box in the question. These cues are used to define Q to detect a question-related subgraph. The QA system uses the question-related subgraphs and the key phrases in the question to guide the answer. If a question contains 'where', the answer may related to the Location class instance. Likewise, a question included 'how' tends to ask about the Task class instances. As shown in the bottom left of Fig. 9, a subgraph related to the question, the reasoning path, can be demonstrated together. For more details, it also demonstrates a corresponding document as in the bottom right of Fig. 9. These evidences facilitate the user's better understanding of the answer.

6.4. Limitations

There are some limitations to our approach as follows.

Performance As mentioned before, we employ a LS-Llama model, leveraging LoRA. It only trains a single layer while a LLM is frozen. Although it enables parameter-efficient fine-tuning, it is unable to

Question Answering System with DoorSystem Knowledge

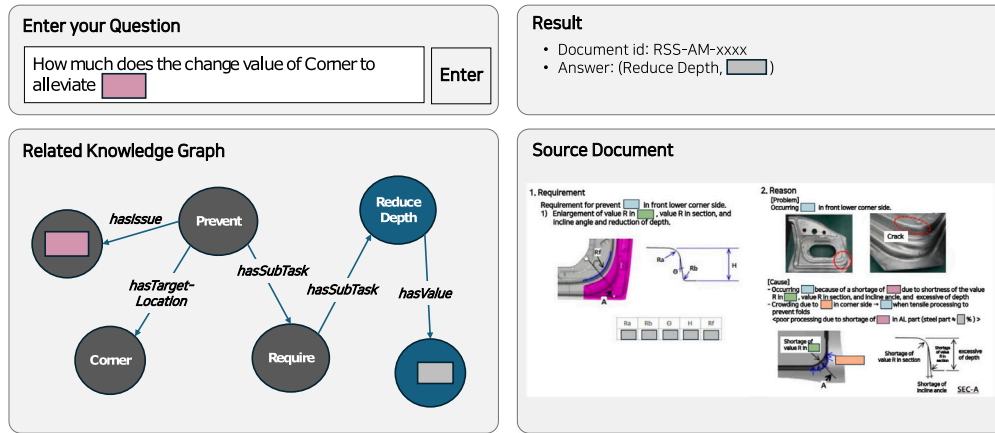


Fig. 9. The web demonstration of QA system with Task-Centric KG.

fully leverage the capabilities of the LLM. Furthermore, our resource constraints pre-vented us from experimenting with LLMs with more than 8 billion parameters, potentially limiting the performance of the proposed model.

Reproducibility The proprietary data used to design TCO and fine-tune OmEGa could not be publicly disclosed due to the security policy of the company that funded this research. This may hinder the reproducibility and further validation of our work.

Adaptation domain Since TCO is derived from the DoorSystem dataset, it focuses on the door system domain of automobiles. Nevertheless, since TCO is designed for real-world automotive manufacturing processes, we believe it can be reused in similar processes related to automotive manufacturing.

Data format OmEGa imports PDF files and extracts the bounding boxes of words for spatial embedding. So documents with other extensions must be converted to PDF format.

Scalability OmEGa extracts information and builds KGs at a document level one by one. Therefore, the number of input documents and the processing time are proportional to each other. However, human feedback is required to correct the errors that may occur after knowledge reasoning. The human-in-the-loop process may be costly.

7. Conclusions

In this work, TCO is designed for the manufacturing domain, enabling the extraction of meaningful information from domain-specific multimodal documents. In OmEGa, spatial embedding, modality linking, and schema-based data augmentation are adopted to mitigate the structural and contextual challenges in understanding unstructured, multimodal, and domain-specific manufacturing documents.

The proposed information extraction framework, OmEGa, effectively transforms unstructured document data into structured symbolic knowledge, demonstrating the applicability through integration with a wide range of disciplines, not just the specialized manufacturing area. Specifically, spatial embedding and modality linking are utilized to mitigate the structural and contextual challenges in understanding unstructured and multimodal documents. By employing parameter-efficient fine-tuning methods on LLMs with schema-based data augmentation, we achieved promising results in domain-specific document understanding. The use case of KGQA demonstrates that KGs built by Omega contain real-world, industry-value knowledge and can enable communication between domain experts.

CRediT authorship contribution statement

Midan Shim: Software, Methodology, Conceptualization. **Hyojun Choi:** Validation, Data curation. **Heeyeon Koo:** Validation, Software. **Kaehyun Um:** Validation, Software. **Kyong-Ho Lee:** Project administration, Conceptualization. **Sanghyun Lee:** Formal analysis, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Kyong-Ho Lee reports financial support was provided by Hyundai Motor Company. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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